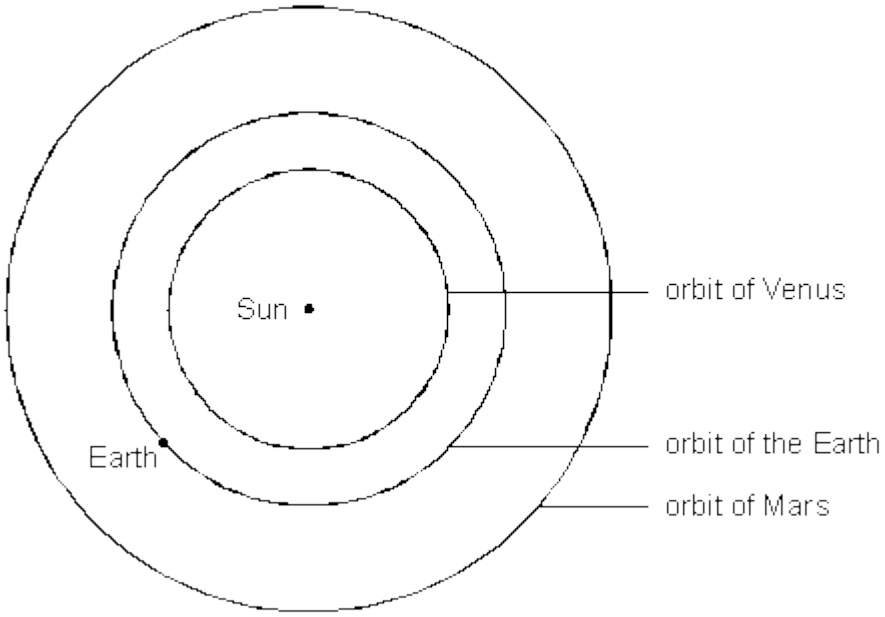
1. The diagram shows the orbits of the Earth, Mars and Venus.
The position of the Earth is shown.



not to scale

A person on the Earth observes Mars and Venus.

(a) (i) On the diagram above, draw **two** more dots to show the positions of Mars and Venus when they are closest to the Earth.

Label the dot for Mars with a letter M and the dot for Venus with a letter V.

1 mark

(ii) Why is it easiest to see Mars when it is closest to the Earth?

… … … … … . . … … … … … … … … … … … … … . … … … …

… … … … … . . … … … … … … … … … … … … … . … … … …

1 mark

(b) What force keeps the Earth in its orbit and stops it flying off into space?

… … … … … . . … … … … … … … … … … … … … . … … … … … ..

1 mark

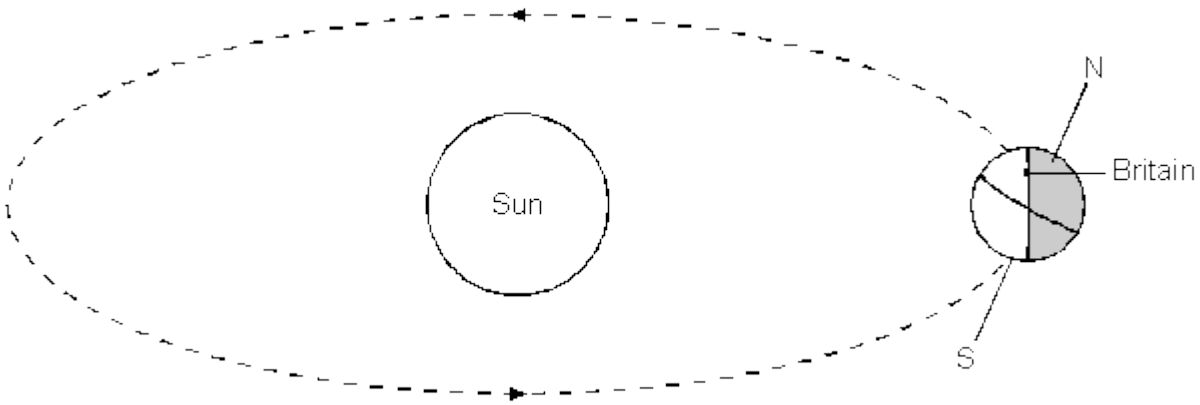
(c) From the Earth, the Moon always looks approximately the same size.
What can you conclude from this about the orbit of the Moon around the Earth?

… … … … … . . … … … … … … … … … … … … … . … … … … … ..

… … … … … . . … … … … … … … … … … … … … . … … … … … ..

1 mark

(d) The diagram shows the Earth in its orbit around the Sun.



not to scale

What season is it in Britain? Explain your answer.

… … … … … . . … … … … … … … … … … … … … … . … … … … … ..

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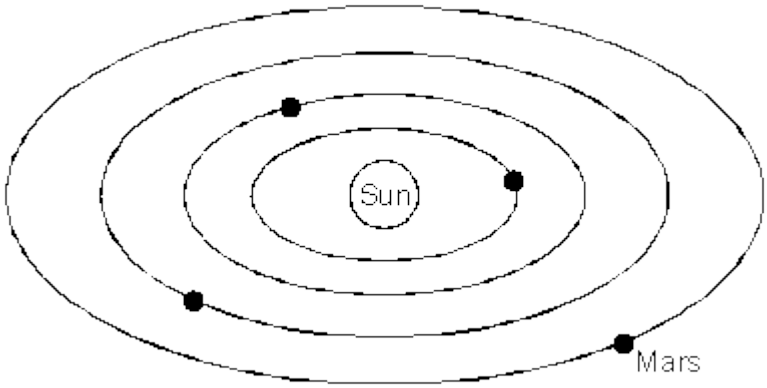
2 marks
Maximum 6 marks

2. Mars is the fourth planet from the Sun.



not to scale

(a) Name **one** planet which is closer than Mars to the Sun.

.....

1 mark

(b) A day and night on Mars lasts nearly 25 Earth hours. Explain why there is daytime and night-time on Mars.

.....

.....

1 mark

(c) Like Earth, Mars has summers and winters.
Suggest why there are seasons on Mars.

.....

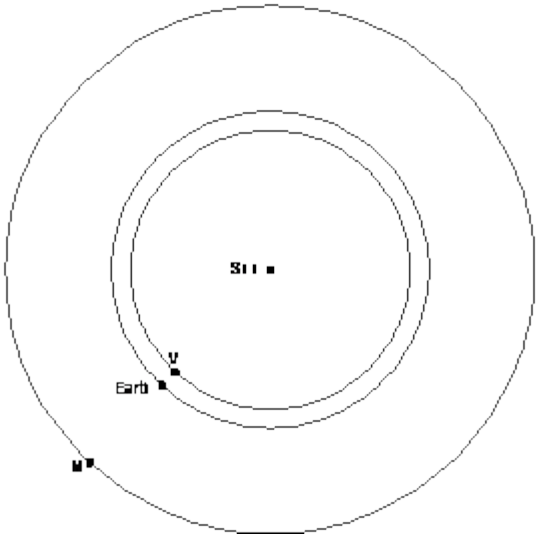
.....

1 mark

Maximum 3 marks

Mark schemes

1. (a) (i) the dots for Mars, the Earth and Venus should be in a straight line, the shortest possible distance apart



1 (L5)

(ii) any **one** from

- it is brightest
  accept 'it is brighter'
- it looks biggest **or** bigger
  accept 'it is bigger'
  accept 'you can see it at night'

1 (L5)

(b) gravity

1 (L5)

(c) any **one** from

- it is circular
- it is always the same distance from the Earth

1 (L6)

(d) winter

1 (L6)

any **one** from

- the Northern hemisphere is tilted away from

  *do **not** accept ‘ the Earth is tilted the Sun away from the Sun’*

- Britain is in the part of the Earth which is tilted away from the Sun

- the Sun’s rays hit the Earth at more of an angle

  accept ‘the Sun appears lower in the sky’ angle

  accept ‘night is longer than day’

1 (L6)

[6]

2. (a) any **one** from

- Mercury
- Venus
- Earth

1 (L5)

(b) it rotates on its axis

*accept ‘it spins’ **or** ‘it turns on its axis’*

***or** ‘it rotates’ **or** ‘it turns’*

accept ‘it spins as it goes round the Sun’

*do **not** accept ‘it spins and it goes round the Sun’*

*do **not** accept ‘it goes round **or** moves round’*

1 (L5)

(c) its axis is tilted

*accept ‘sometimes one pole is closer to the Sun and sometimes the other is’ **or** ‘sometimes one pole is in daylight’*

1 (L6)

[3]